



Air University, Islamabad

Project Proposal

CyberGuard AI: AI-Driven Cyber Risk Analysis and Compliance Monitoring System

Version 2.0

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Project Category

☐ **A** - Desktop Application/Information System ✓ **B** - Web Application/Web Application based Information System ✓ **C** - Problem Solving and Artificial Intelligence ✓ **D** - Simulation and Modeling ☐ **E** - Smartphone Application ☐ **F** - Smartphone Game

Abstract

CyberGuard AI is an AI-driven cyber risk analysis and compliance monitoring system developed to incorporate enterprise security by combining ML risk inspection, real-time monitoring, and explainable AI into a single platform. Current systems face cybersecurity threats that arise from system misconfigurations, security flaw, and non-compliance with the standard benchmarks by Center for Internet Security (CIS). There are some similar platforms like Microsoft Defender for Cloud that offers cloud-based compliance monitoring, and Tenable.io gives agentless scanning through cloud APIs, but there is no single platform that combines configuration examination, ML-driven anomaly detection, attack path simulation, and XAI integration. CyberGuard AI proposes an enhanced system, that uses an agentless multi-OS architecture, supervised and unsupervised ML models for anomaly detection, MITRE ATT&CK-informed lateral attack movement simulation, XAI incorporation to help analysts understand the root cause behind every alert, hence reducing alert fatigue, and LLM integration. Security analysts can derive AI-generated insights and take actions accordingly. CyberGuard AI represents a step towards action-oriented, self-improving and analyst-friendly security operations.

Introduction

In the modern world, majority of the enterprises and organizations face cybersecurity related problems. As a result, the outlook of cybersecurity has become more complex because of internal misconfigurations, digital attacks, and drift from compliant behaviour. As organizations expand their infrastructure to Windows and Linux systems, IoT endpoints, firewalls, maintaining a compliant and secured network at a large scale has become difficult.

Existing systems address this problem but individually they do not work as good. Vulnerability scanners like Tenable Nessus and Qualys VMDR provide Center for Internet Security(CIS) benchmark compliance checking, and platforms like Tenable.io offers agentless scanning via SSH and cloud APIs, and Microsoft Defender for Cloud extends compliance checking to cloud-based environments. However, these systems have their own set of limitations. Alerts are rule-based and there is no use of ML in order to adapt to specific environments over time. Agentless architecture leads to gaps like access restrictions and vendor API dependencies. Moreover, no existing system combines ML based configuration auditing, attack path simulation, natural language querying, XAI incorporation to reduce alert fatigue, and LLM-generated insights across the enterprise infrastructure [1].

CyberGuard AI is designed to bridge these gaps by incorporating an AI/ML intelligence layer on top of conventional compliance systems. This agentless system will collect configuration data across multiple operating systems, and evaluating it against CIS benchmarks, in addition to detecting drifts using supervised/unsupervised machine learning models. Deep neural networks perform analysis to check for any changes in the series of configuration over time. The system aims to monitor real-time events and prioritize the responses based on scores. Explainable AI and natural language querying using Large Language Models (LLMs) are a part of the system so that every alert is followed by an explainable justification which is understandable by a lay-person. The system supports a MITRE ATT&CK path simulation that gives an idea about how the risk/attack could spread through an organization's framework.

Problem Statement

Many organisations today deal with an ever-growing number of cybersecurity incidents, which may be linked to issues like misconfigured systems, outdated software's or just deviation from established security practices. CIS, the Center for Internet Security, publishes detailed benchmarks on how to secure environments like Windows, Linux or network devices, but even such thoroughly defined frameworks as CIS require organisations/industries to comply with them manually. Such a process is slow, inconsistent and hard to scale.

Statistics clearly show how serious of an issue this is. According to Verizon's 2023 data breach investigations report, misconfigurations are responsible for 30% of error-related breaches[2]. XM cyber analysed around 40 million security exposures and deduced that 80% of them occur due to identity and credential misconfigurations, one-third of which directly put critical assets at risk[3]. Despite such alarming figures, most organisations lack a system that could detect vulnerabilities, evaluate how they relate to each other and forecast possible threats.

Existing tools do parts of such a job. Tenable Nessus and Qualys VMDR are popular

solutions used for vulnerability scanning and CIS benchmarking. Splunk and Microsoft Sentinel offer capabilities of event monitoring and alerting. But, these tools were not designed to work together, and each has its own limitation and flaw. An important observation is that these solutions rely on their alerts being rule-based, i.e. they flag what they already know to look for, and lack the ability to learn from a particular environment.

They also have no way to stimulate what an attacker would do or how they would behave once they're inside the network. Whether a compromised node is a dead end or a direct path to a critical system like a database server makes a significant difference in calculating the actual risk, but existing tools make no such distinction.

Then there is an issue of alert fatigue. Security teams receive multiple alerts from different sources, but as there are no explanations, analysts spend a lot of time deciphering the anomalies rather than fixing them. This is the gap where a lot of risk lies between detection and understanding.

Proposed Solution

CyberGuard AI is a system that is built on the idea that compliance and threat detection must be done in parallel. It takes into account the capabilities of existing tools such as configuration auditing against CIS benchmarks, and adds a layer of AI/ML functionalities for automation and intelligence. This makes the results more actionable rather than giving away information.

The system uses an agentless approach to gather configurations data for supported devices including Windows and Linux, without the need to install any software on monitored endpoints. All collected data has metadata fields like timestamps and versions. This is used to create historical baselines for every operating system and host groups. By comparing current configurations with such baselines, we enable the discovery of any inconsistency which would go unnoticed between scheduled scans.

For detection, the system makes use of both supervised and unsupervised ML models. Supervised models deal with patterns that are already known to exist, while unsupervised flags configurations that fall outside the learned norm, despite not having been occurred previously. A sequential deep learning model analyzes configuration changes over time, and detects any unusual activity/suspicious pattern that only become visible when we look at a sequence of changes rather than a single snapshot.

The system implements an attack path simulation based on MITRE ATT&CK techniques. If in any case, a device is flagged as compromised or misconfigured, the system predicts how the attacker can exploit that asset and move through the network from that point, providing analysts with a clearer picture of actual risk, not just a list of findings. Each alert is then accompanied with an XAI (explainable AI) breakdown, identifying which configuration fields are responsible for generating such an alert, which ensures that analysts do not have to sift through raw data to get the information they need. An LLM-based natural language interface enables analysts to query the system in plain English, for example, asking for the top misconfigurations from the past week across all Linux hosts.

Objectives

- BO-1: Collect and version configuration baselines across multiple OS types using an agent-less architecture
- BO-2: Identify both common and unique misconfigurations with the use of supervised and unsupervised machine learning algorithms
- BO-3: Identify suspicious configuration change sequences through time-aware deep learning analysis
- BO-4: Evaluate the risk of lateral movement based on MITRE ATT&CK attack simulation
- BO-5: Decrease alert fatigue through explainable AI and prioritization through machine learning
- BO-6: Enable natural language querying of compliance posture through LLM integration

Related System Analysis/Literature Review

The research conducted by (Rastogi et al., 2025) consisted of a mixed-methods approach, stretching to a total of 3 months[1]. It involved surveying around 248 security professionals, and conducting interviews of a total of 24 analysts, examining how they process and trust AI generated explanations. One important thing we learned from this study is that choosing the appropriate level and format of explanation in environments demanding rapid decision making under high stakes conditions remains a challenging task. A key finding was that analysts at different tiers need different forms of explanations, with Tier-1 analysts requiring concise explanations when sorting alerts, while Tier-2 analysts required richer, and more narrative-driven explanations. Currently available tools often generate inadequate explanations that fail to adapt to the analyst’s situations, possibly leading to alert fatigue. This is directly relevant to CyberGuard AI’s XAI module, which aims to deliver per-alert feature-level explanations according every to analyst’s role and needs. This will avoid generic, one-size-fit-all outputs.

A paper published by (Wang and Yang, 2025) on arXiv, titled “Machine Learning-Based Cloud Computing Compliance Process Automation” introduced a ML framework designed especially for the automatic assessment of cloud compliance processes [4]. This strongly relates to CyberGuard AI’s goal concerning configuration auditing. The model proposed in the paper includes the use of BERT for processing documents, One-class SVM for detecting anomalies, and a CNN-LSTM architecture for sequential compliance data analysis, leading to compliance process time being reduced from 7 days to 1.5 days and manual labour costs lowered by 73%. Although the model isn’t focused on evaluating OS-level configuration benchmarks like CyberGuard, it proves the feasibility and effectiveness of using hybrid ML models for compliance automation tasks. CyberGuard AI extends this along the same lines by implementing a similar approach to the assessment of CIS benchmarks on Windows and Linux, with an extra consideration of baseline versions.

Sholihin and Salman [5] developed a tool for automating audits named OSCAT (One Stop CIS-Benchmarking Auditing Tool), tested against 248 security controls on Red Hat Enterprise Linux 7 using CIS benchmark version 3.1.1. The results showed that OSCAT efficiently categorized audit results into Pass, Fail and manual review. OSCAT had clear advantages in terms of speed and consistency compared to manual methods, reducing the manual workload and increasing efficiency in identifying and managing security risks. Although OSCAT proves that CIS benchmarking is automatable, but it was only done for one particular OS, and no Machine Learning was implemented in order to detect deviations. CyberGuard takes this approach further by extending the automated CIS compliance checking to windows, and also adding baseline versioning and ML-based anomaly detection that goes beyond static pass/fail outputs.

A systemic literature review published in 2025 on arXiv analysed 17 different studies (research papers) on security assessments using CIS benchmarks between 2017 and 2025 [6]. The review discovered a trend towards automation and continuous CIS compliance, but it also pointed out that current efforts only concentrate on auditing and enforcement, paying little to no attention towards visualizing the results. This gap is filled by CyberGuard AI's dashboard module, which uses risk scored visuals and heatmaps to display compliance and deviation data, rather than raw audit outputs. GraphSecure analyzes the security of AWS accounts by mapping them against CIS benchmarks. This tool can work with multiple AWS accounts, additionally providing recommendations on how to deal with failed benchmarks. GraphSecure provides a visualization of scanned vulnerabilities in a system in the form of chart, helping users review their account's security status. SIEM (Security Information and Event Management) based systems deliver deep insights into cybersecurity challenges. As cyber threats grow and become complex, AI based management has become essential in systems. Machine Learning significantly improves conventional SIEM functionalities by reducing false positives and addressing the limitations of rule-based systems [7].

Application of CIS benchmarks improves security profiles of Ubuntu Server 22.04 by reducing vulnerabilities and making system resilient against cyber attacks [8]. Broader implementation of CIS standards across multiple operating systems is gap that is covered by CyberGuard AI, that integrates automated compliance checking and continuous monitoring, leading to more developed and adaptive systems.

In the paper, Agentless Endpoint Security Monitoring Framework[9] by Asem Ghaleb, our benchmark inspiration for our proposed solution, the writer proposes a very intuitive system architecture similar to our own. This consists of a controller section which is a management service that manages and establishes connections to the remote devices. The second part is a retrieval agent responsible for collecting system configuration from host. This can be done by storing the target machine configuration in to a set of .conf files. The workflow is that the framework server maintains a list of host addresses in host.conf file, it then reads rawdata.conf that defines the data our agentless system must read from host. Once the SSH connection is established, it sends a list of commands for execution on target machines to pull configuration and saving them in the form of json format in any NoSQL Database. This centralized database is defined as the data repository in the paper[9].

The fourth part of this architecture is the use of security models which differentiates

between normal configuration from suspicious ones. The authors implemented a logistic regression model that achieved an accuracy of 83% accuracy. Finally, a reporting module is proposed in which all the key findings and reports are displayed to the stakeholder. This elongates with our architecture we propose. The improvements we propose is the integration of a benchmarks datasets derived from CIS benchmarks as a ground truth. Furthermore, CyberGuard intends to visualize the attack path in the network that a malicious attack will take to spread to the network.

In addition, Ahmed et al[10] proposes a deep learning based intrusion and anomaly detection framework for cloud data center security. The authors evaluate the results of multiple deep learning models such as RNNs, LSTMs, CNNs trained on benchmark datasets such as NSL-KDD and UNSW-NB15. Their study focuses on identifying malicious attack patterns in cloud infrastructure by using supervised learning models. Their experiments showed that LSTMs achieved the highest ROC-AUC score of 0.9. This metric outperforms the RNN and CNN models in terms of predicting persistent and time dependent attack patterns. However, the study primarily focuses on the model performance rather than system level architecture or deployment mechanisms such as agentless data collection. This work supports our proposed system by providing benchmark deep learning models for anomaly detection that can be added to our architecture for improved threat classification and predictive security analytics.

Comparison of Existing Systems and What CyberGuard AI Adds

Existing system	What it does	Gap	What CyberGuard AI adds
Tenable Nessus	Scans systems against known CVEs and CIS benchmarks; generates compliance reports	Rule-based, reactive detection only; no ML, no temporal analysis, no attack path modeling	ML anomaly detection, sequential pattern analysis, MITRE ATT&CK attack path simulation
Qualys VMDR	Agentless vulnerability scanning, asset inventory, compliance checks across cloud and on-premise	No predictive capability; alerts lack explainability; cannot simulate attacker lateral movement	XAI-driven alert explanations, predictive compliance forecasting, lateral movement modeling
Splunk (SIEM)	Real-time log ingestion, event correlation, rule-based alerting across enterprise systems	Requires significant manual tuning; high false positive rates; no configuration baseline versioning	Automated baseline versioning per OS, ML-based alert prioritization, reduced false positives via feedback loop
Microsoft Sentinel	Cloud-native SIEM with pre-built ML detections, threat intelligence integration, and SOAR capabilities	ML detections are pre-built and not customizable to specific configuration baselines; no CIS benchmark compliance	Custom ML models trained on organization-specific baselines, CIS-aligned compliance auditing, agentless multi-OS coverage
CrowdStrike Falcon	Endpoint detection and response (EDR), behavioral analysis, threat hunting	Agent-required deployment; limited coverage for firewalls, IoT, and legacy devices; no compliance reporting	Agentless architecture for Windows and Linux, CIS compliance monitoring, LLM-based natural language querying

Vision Statement

For security analysts and compliance officers who need to perform continuous configuration audit, check for compliance drift, and respond to cyber threats promptly across the enterprise infrastructure, CyberGuard AI is an AI-driven cyber risk analysis and compliance monitoring system that combines agentless multi-OS configuration auditing, ML-based anomaly detection, MITRE ATT&CK-informed path simulation, and LLM-based natural language querying and insight generation. Unlike existing platforms such as Tenable Nessus, Qualys VMDR, and Microsoft Defender for Cloud that perform compliance checking, scanning for vulnerabilities, and event monitoring whilst relying on rule-based alerts and lacking an intelligent explainable component, CyberGuard AI offers security analysis with explainable alerts, prioritized and actionable insights that reduce alert fatigue, enabling proactive defense across devices on an agentless architecture.

Scope

The scope of the proposed project focuses upon developing an intelligent, scalable, agentless platform that seeks to evolve traditional cyber monitoring systems by integrating AI focused capabilities for risk detection and compliance monitoring. The system is designed to operate on multiple devices and operating systems. It will collect and manage configurations and event data in order to establish security baselines against CIS benchmarks and detect inconsistencies with standard practices. Its main features include

1. Secure user authentication with role based access
2. Automated data collection and ingestion by following the principle of least privilege avoiding unnecessary administrative access to monitored systems.
3. Protecting collected data through encryption during transmission and storage while maintaining data integrity and confidentiality.
4. Baseline configuration management
5. AI based anomaly detection using both supervised and unsupervised learning methods

Additionally, the system performs real time monitoring, assigns risk scores to anomalies. It also generates alerts with explanations to assist analysts in decision making. The platform also supports real time monitoring, dynamic network mapping to identify the path a potential attack can use to move over the network, and predictive compliance analysis. It also provides an interactive web dashboard. Overall, the scope is mainly limited to detection, monitoring, and decision support features. The scope does not include automated actions against threats. It emphasizes modularity, scalability and flexibility to accommodate future needs

Project Modules

Module 1: Authentication and User Management

- **FE.1:** Login/Sign-In with Session handling and password hashing
- **FE.2:** Multi factor authentication system enforced for critical roles.
- **FE.3:** Session Management (token generated for user login and sign-in)
- **FE.4:** Oauth and SSO Integration
- **FE.5:** Password reset and recovery system
- **FE.6:** Role based access control for admin and analyst

Module 2. Configuration Collection and Data Ingestion

- **FE.1:** Agentless architecture: Multi-OS configuration
- **FE.2:** Scheduled and On-demand Scanning for collecting updated data
- **FE.3:** Incremental data collection
- **FE.4:** Secure Transport Layer for Data Encryption

Module 3. Baseline and Data Management

- **FE.1:** Automatic Configuration Profile Generation
- **FE.2:** Importing Industry Standard Security configurations
- **FE.3:** Custom Rule Definition
- **FE.4:** Baseline Versioning and Configuration Rollback
- **FE.5:** Dataset Labeling for ML Training
- **FE.6:** Synthetic Data Generation to enhance the training process

Module 4. AI/ML anomaly Detection Engine

- **FE.1:** Unsupervised Anomaly Detection
- **FE.2:** Supervised Classification
- **FE.3:** Temporal Analysis
- **FE.4:** Feature Engineering
- **FE.5:** Model Retraining and Feedback loop
- **FE.6:** Explainable AI (XAI)

Module 5: Real-Time AI-Powered Monitoring & Event Stream Analysis

- **FE.1:** Continuous Event Collection
- **FE.2:** Machine Learning-Based Anomaly Detection
- **FE.3:** AI-Based Event Correlation Engine
- **FE.4:** Risk Scoring and Alert Prioritization

Module 6: Alerting, Visualization Decision Support System

- **FE.1:** AI-Based Alert Prioritization
- **FE.2:** Intelligent Alert Suppression and False Positive Reduction
- **FE.3:** Multi-Channel Smart Notifications
- **FE.4:** Explainable AI (XAI) for Alert Transparency
- **FE.5:** Predictive Risk and Compliance Insights
- **FE.6:** AI-Powered Visualization and Executive Insights

Module 7: AI-Powered Network Mapping & Threat Propagation Modeling

- **FE.1:** AI-Based Asset Discovery and Identification
- **FE.2:** Dynamic Network Topology Mapping
- **FE.3:** AI-Driven Lateral Movement Simulation
- **FE.4:** Intelligent Segmentation and Firewall Optimization
- **FE.5:** Unauthorized Device and Network Change Detection
- **FE.6:** Automated Containment and Isolation Recommendations

Module 8: System Administration Settings

- **FE.1:** Target Host Management
- **FE.2:** ML Model Management
- **FE.3:** Platform Health Monitoring
- **FE.4:** Audit Logs and Administrative Activity Tracking
- **FE.5:** AI Feedback Integration from Security Analysts

System Limitations/Constraints

No Automated Remediation

The system does not perform automatic mitigation or fixing of detected vulnerabilities or misconfigurations. It only detects, alerts and explains why an alert was generated.

No Guarantee of Complete Threat Detection

The system cannot guarantee detection of all cyber threats, especially highly sophisticated or relative new attacks

Limited Depth of Data Collection (Agentless Constraint)

The system relies on remote protocols and cannot access low level details such as kernel level activities or real time system calls that typically require an agent.

No Real-Time Continuous Monitoring at Host Level

The system uses scheduled and on demand monitoring. It cannot provide true continuous, real time monitoring of all activities occurring between time intervals

Inability to Monitor Systems Without Remote Access

The system only supports monitoring devices that allow remote access. Devices behind restricted firewalls or with disabled SSH/WinRM cannot be monitored by the system.

Data Gathering Approach

This project's requirement gathering was conducted through stakeholder meetings, backed by expert consultation, and competitive analysis. The project's team conducted online meetings with the industrial supervisor and refined the project modules based on industry expectations. The finalized modules were further validated and approved by the university supervisor, to align with the academic requirements and project objectives. Furthermore, a comparative analysis was conducted of existing platforms in order to identify their limitations which were the foundational capabilities of CyberGuard AI.

Tools and Technologies

Tools and Technologies	Category	Tools
Frontend	Frontend Framework	React.js (Web-based User Interface)
	Visualization	D3.js / Chart.js for Dashboards and Reporting
Backend & Database	Backend Framework	Django or FastAPI (Python-based Web Framework)
	Database	PostgreSQL (Relational Database Management System)
Continued on next page		

Tools and Technologies	Category	Tools
AI & ML	Machine Learning	Scikit-learn, TensorFlow, PyTorch
	Explainable AI	SHAP, LIME
	LLM Integration	OpenAI API, LangChain
Security & Compliance	Configuration Collection	SSH, WinRM (Agentless Configuration Collection)
	Security Benchmarks	CIS Benchmarks, MITRE ATT&CK Framework
Development & Infrastructure	Version Control	Git, GitHub
	Deployment	Local Deployment in OT Environment

Tools and Technologies for CyberGuard AI

Project Stakeholders and Roles

Stakeholder	Role and Responsibility
Zara Zaman, Malik Saad Ahmed, Liza Asad	Designing, development, testing, and documentation
Dr. Madiha Yousaf	University Supervisor, project evaluation, academic mentoring
Mr. Fahad Ahmed	Feedback on proposed modules, alignment with industry expectations
Final Year Project Committee	Project evaluation and complete assessment

Stakeholders of CyberGuard AI

Module based Work Division

Student Name	Roll No	Work Division
Zara Zaman	231248	Module 1 (Auth & User Mgmt), Module 4 (AI/ML Anomaly Detection), Module 7 (Network Mapping & Threat Modeling)
Liza Asad	231166	Module 2 (Configuration Collection & Data Ingestion), Module 5 (Real-Time Monitoring & Event Analysis), Module 8 (System Administration Settings)
Malik Saad Ahmed	231224	Module 3 (Baseline & Data Management), Module 6 (Alerting & Visualization DSS), Module 7 (Network Mapping & Threat Modeling)

Module Based Division for CyberGuard AI

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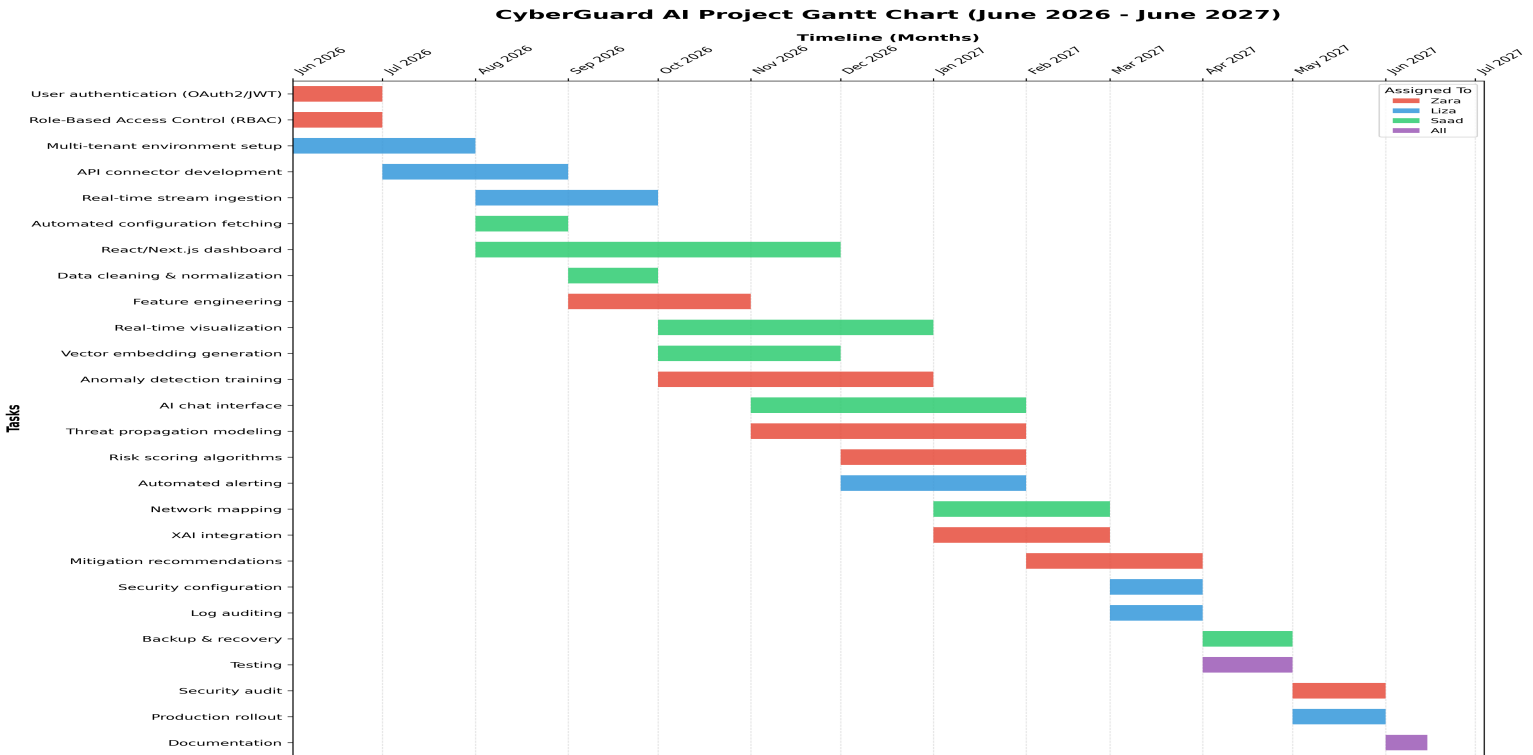
Appendix 1.0

WBS and Gantt Chart

WBS ID	Phase / Major Task	Sub-Tasks & Deliverables	Start	End	Priority	Allocated To
1.0	Project Initiation	User authentication (OAuth2/JWT)	Jun 2026	Jul 2026	High	Zara
		Role-Based Access Control (RBAC)	Jun 2026	Jul 2026	High	Zara
		Multi-tenant environment setup	Jun 2026	Aug 2026	Medium	Liza
2.0	Data Ingestion	API connector development	Jul 2026	Sep 2026	High	Liza
		Real-time stream ingestion	Aug 2026	Oct 2026	High	Liza
		Automated configuration fetching	Aug 2026	Sep 2026	Medium	Saad
3.0	Data Preprocessing	Data cleaning & normalization	Sep 2026	Oct 2026	High	Saad
		Feature engineering for risk metrics	Sep 2026	Nov 2026	High	Zara
		Vector embedding generation	Oct 2026	Dec 2026	Medium	Saad
4.0	AI/ML Core Engine	Anomaly detection model training	Oct 2026	Jan 2027	High	Zara
		Threat propagation modeling	Nov 2026	Feb 2027	High	Zara
		Cyber risk scoring algorithms	Dec 2026	Feb 2027	High	Zara
		Explainable AI (XAI) integration	Jan 2027	Mar 2027	Medium	Zara
5.0	Frontend Development	Responsive React/Next.js dashboard	Aug 2026	Dec 2026	High	Saad
		Real-time visualization components	Oct 2026	Jan 2027	High	Saad
		AI interactive chat interface	Nov 2026	Feb 2027	Medium	Saad
6.0	Decision Support System	Automated alerting & notifications	Dec 2026	Feb 2027	High	Liza
		Network mapping visualization	Jan 2027	Mar 2027	High	Saad
		Mitigation recommendations	Feb 2027	Apr 2027	Medium	Zara
7.0	System Administration	Security settings configuration	Mar 2027	Apr 2027	High	Liza
		Log auditing & health monitoring	Mar 2027	Apr 2027	Medium	Liza
		Backup & recovery protocols	Apr 2027	May 2027	Low	Saad
8.0	QA & Deployment	Unit & integration testing	Apr 2027	May 2027	High	All
		Penetration testing & security audit	May 2027	Jun 2027	High	Zara

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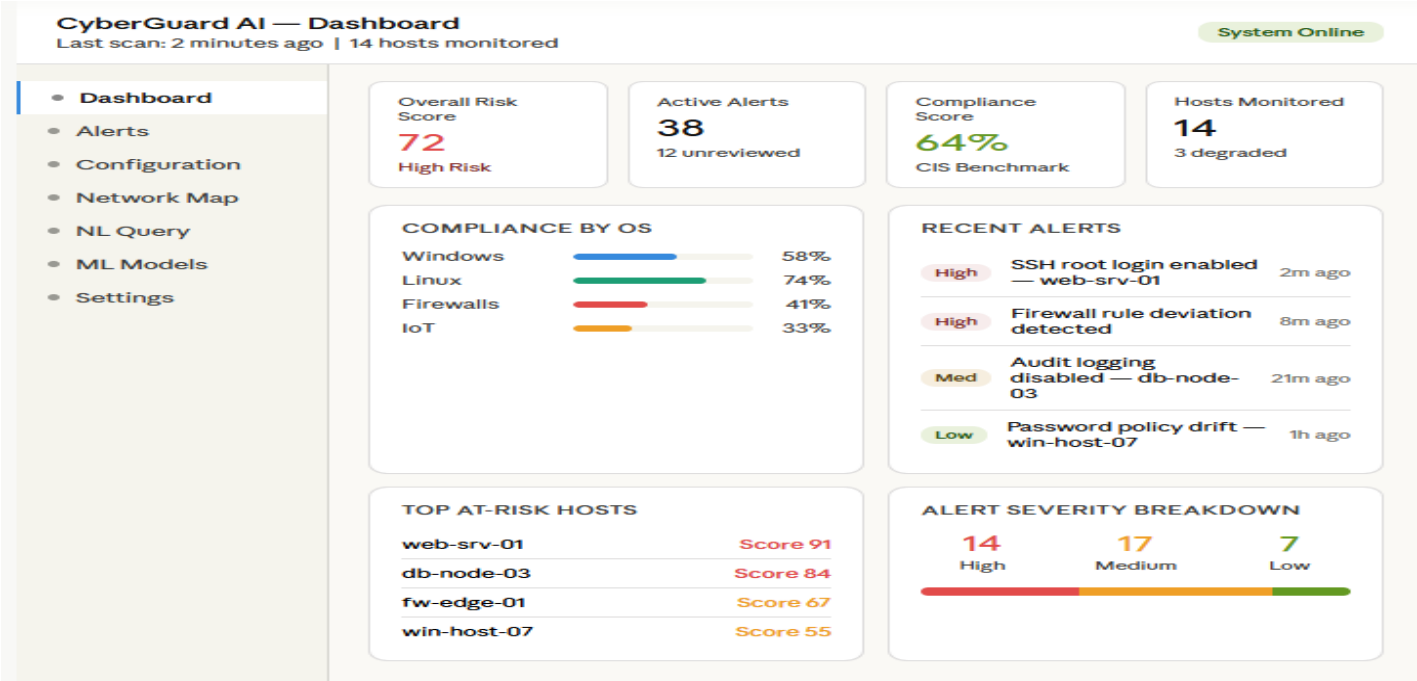
WBS ID	Phase / Major Task	Sub-Tasks & Deliverables	Start	End	Priority	Allocated To
		Production environment rollout	May 2027	Jun 2027	High	Liza
		Final documentation & handover	Jun 2027	Jun 2027	Medium	All
End of Work Breakdown Structure						



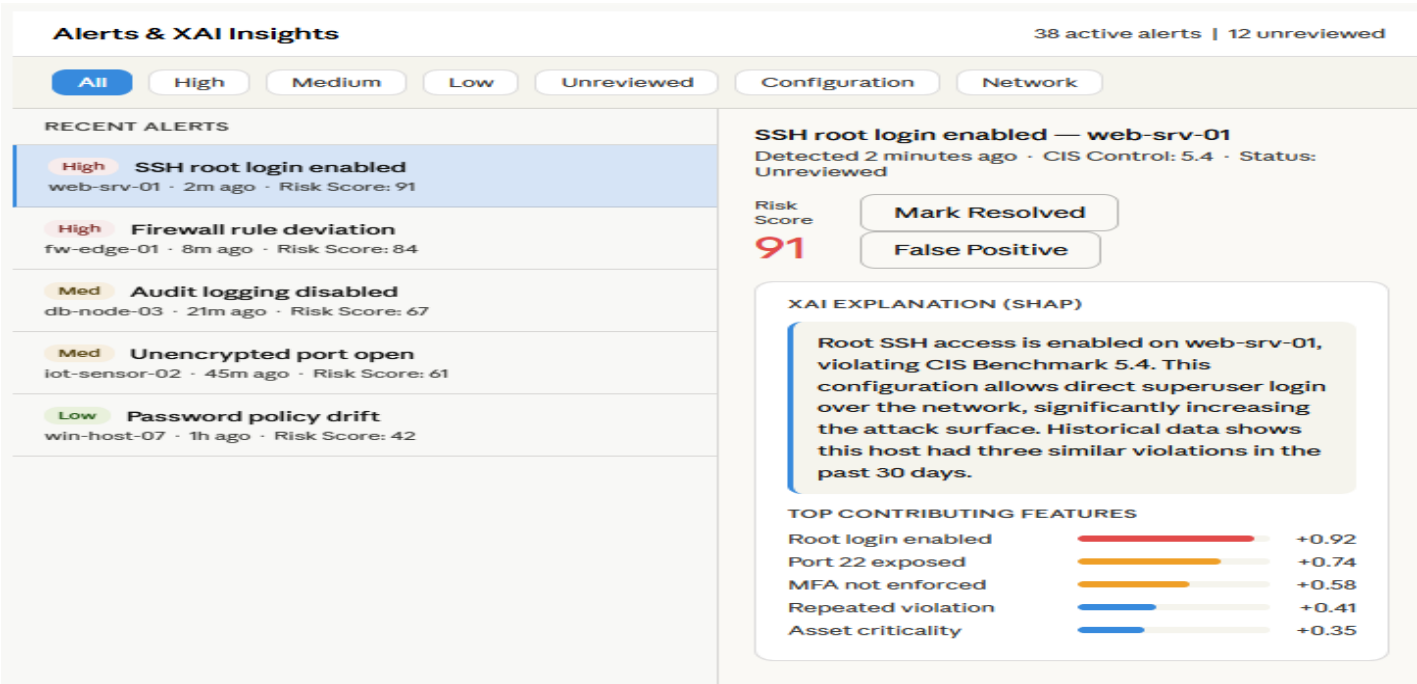
Gantt Chart - CyberGuard AI Project Timeline (June 2026 - June 2027)

Appendix 2.0

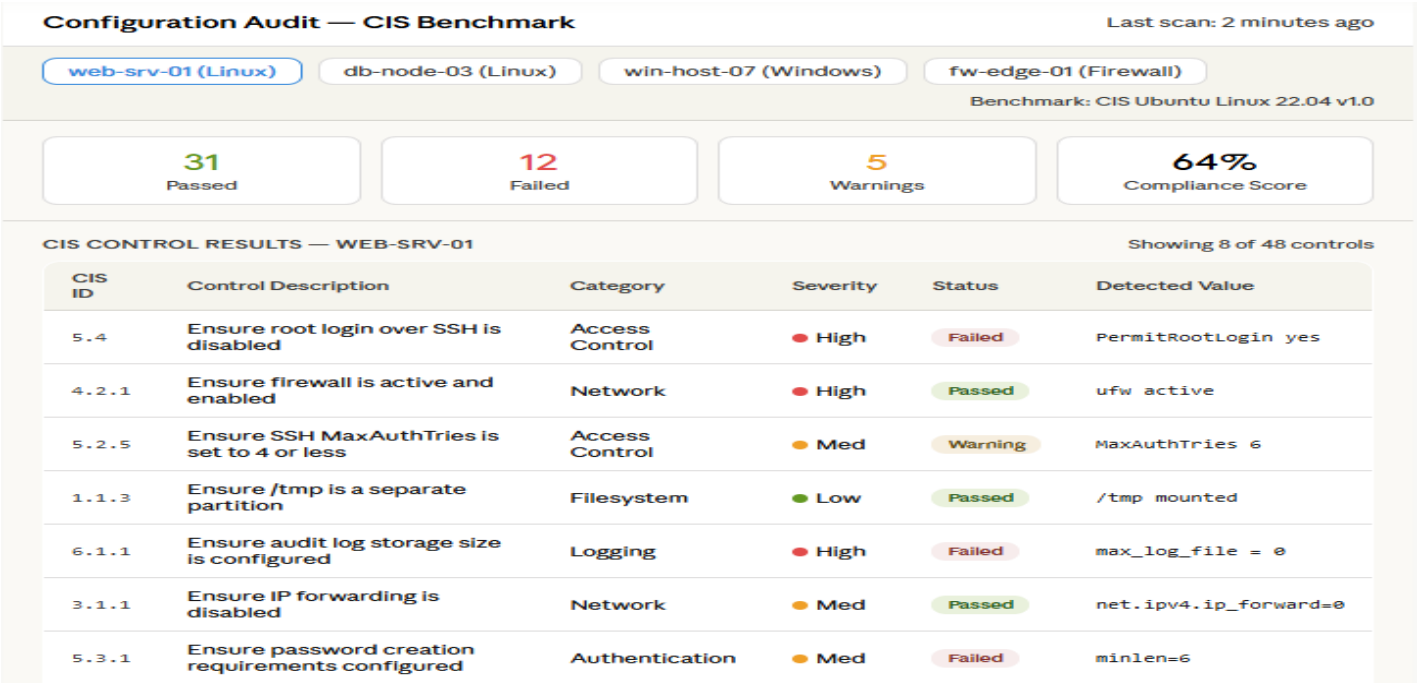
Mockups



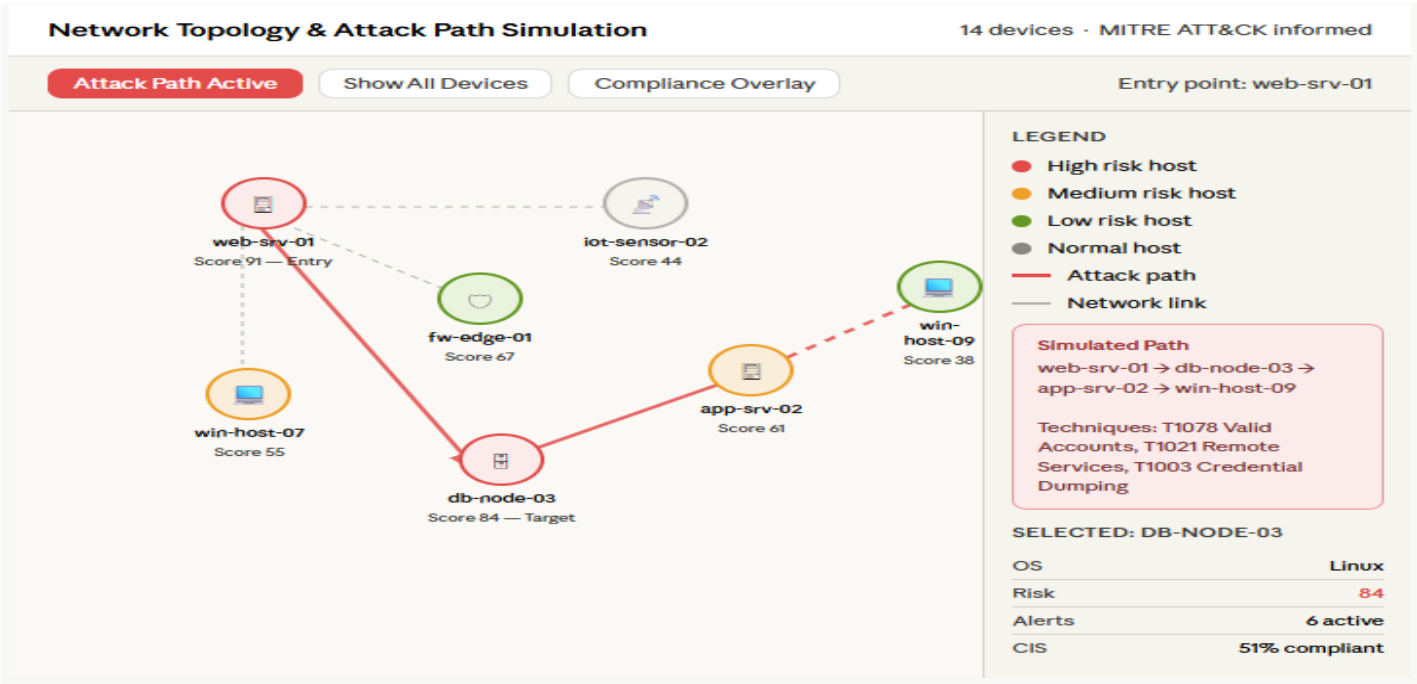
Mockup 1 - Dashboard Overview



Mockup 2 - Alerts and XAI Panel



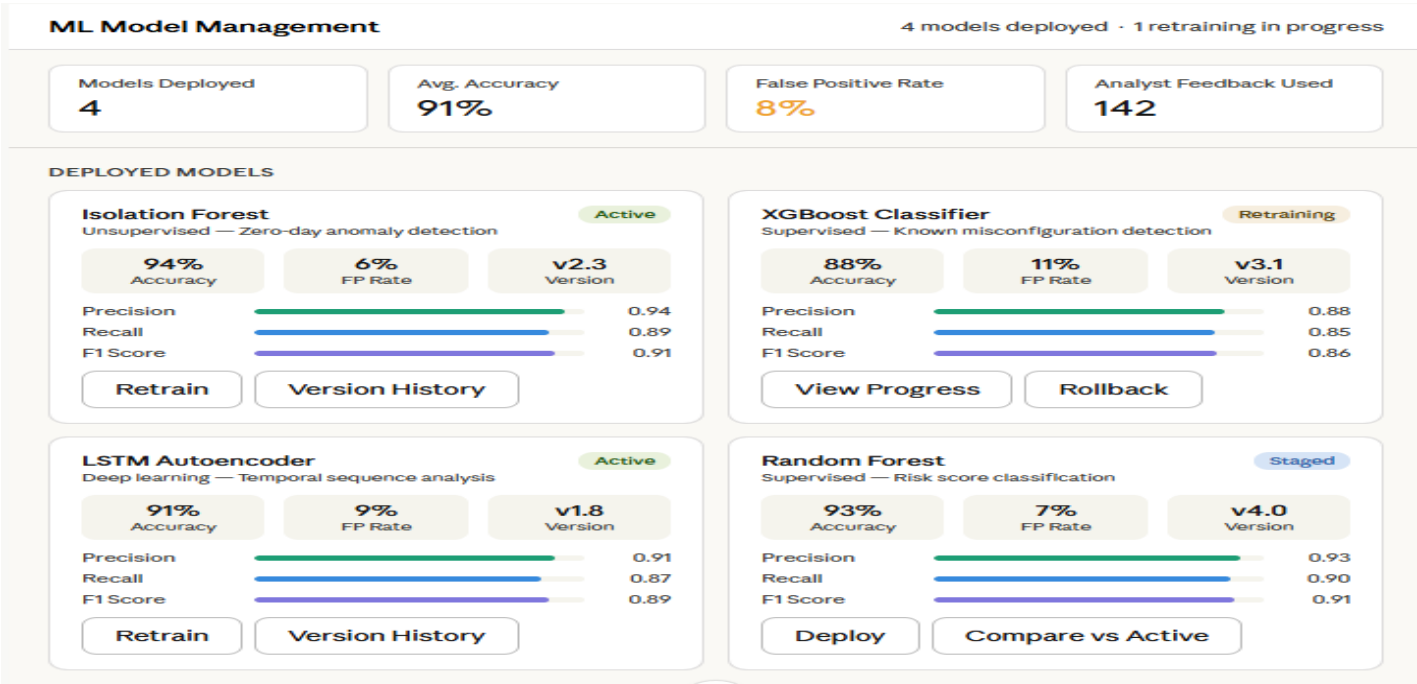
Mockup 3 - Configuration Audit View



Mockup 4 - Network Topology and Attack Path Simulation



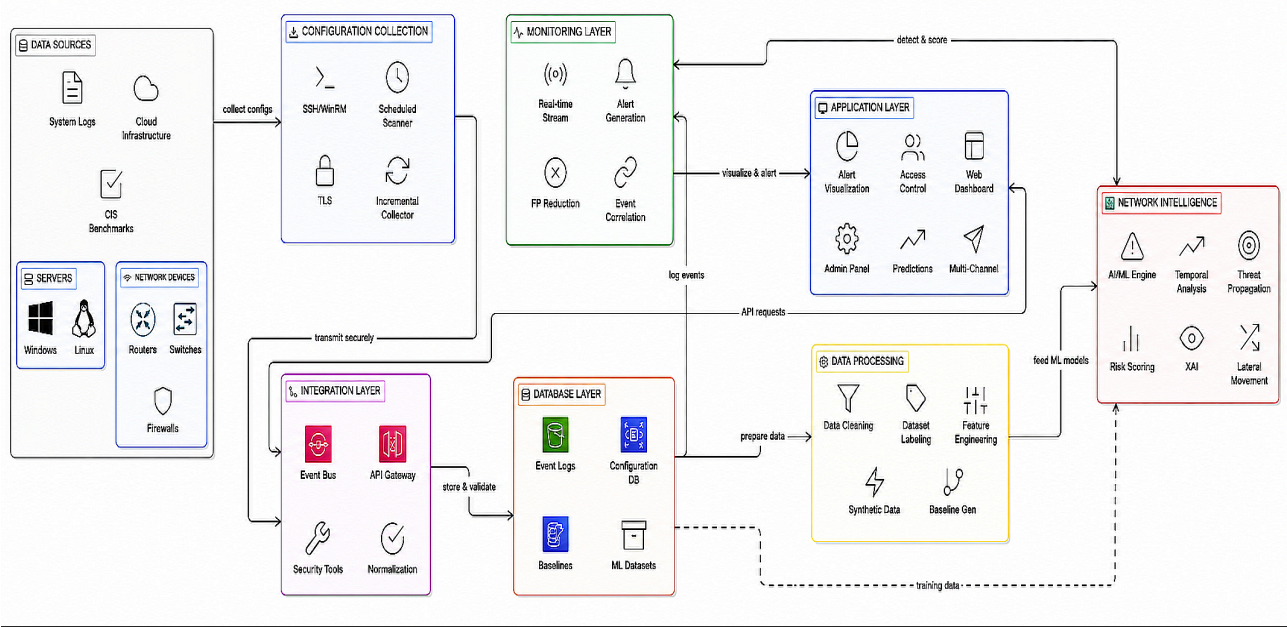
Mockup 5 - ML Model Management



Mockup 6 - Network Mapping and Threat Modeling

Appendix 3.0

Architecture Diagram



Architecture Diagram of CyberGuard AI

Appendix 4.0

Plagiarism Report

Page 2 of 22 - Integrity Overview

Submission ID: trn:oid::3618:137597765

6% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

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- Bibliography

Match Groups

- 16 Not Cited or Quoted** 5%
Matches with neither in-text citation nor quotation marks
- 1 Missing Quotations** 0%
Matches that are still very similar to source material
- 0 Missing Citation** 0%
Matches that have quotation marks, but no in-text citation
- 1 Cited and Quoted** 0%
Matches with in-text citation present, but no quotation marks

Top Sources

- 5% Internet sources
- 0% Publications
- 5% Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

Plagiarism Report

*% detected as AI

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Ai Content Report